# CST8917
# Serverless Applications

## Course Outline

2025-2026

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|---|---|
| **Pre-requisite(s)** | N/A |
| **Co-requisite(s)** | N/A |
| **Prepared by** | John Dallas |
| **Approved by** | John Dallas, Chair |
| **Approval date** | Thursday, December 18, 2025 |
| **Normative hours** | 56.00 |
| **Grading system** | A+ Through F |

| Applicable Program(s) | Level | Core/Elective |
|---|---|---|
| 1531X01FWO - Cloud Development and Operations | 2 | Core |
| 1531X03FWO - Cloud Development and Operations | 2 | Core |
| 1531X07PAO - Cloud Development and Operations | 2 | Core |
| 1531X09FAO - Cloud Development and Operations | 2 | Core |

## Land Acknowledgement

*Algonquin College campuses in Ottawa, Perth and Pembroke are located on the traditional unceded, and unsurrendered territory of the Anishinàbe Algonquin People. The Algonquin People have inhabited and cared for these lands since time immemorial. We take this time to express our gratitude and respect to them and to the land for all that it has provided and will continue to provide.*

*As a post-secondary institution, we acknowledge the harms done to Indigenous Peoples and are committed to learning from the past. We pledge to promote healing and resilience as we move forward in partnership with the Algonquin Nations, First Nations, Métis, and Inuit Peoples in a spirit of reconciliation.*

*While we recognize that territorial acknowledgements are only one step in cultivating greater respect and inclusion of Indigenous Peoples, we commit to accompanying these words with actions. We are dedicated to building a future and community that is better for all.*

*We pledge to continue exploring and making meaningful contributions to the Truth and Reconciliation Commission of Canada's Calls to Action.*

*Learn more about the College's commitment to Truth and Reconciliation:* https://www.algonquincollege.com/tri

## Course Description

The evolution of managed cloud services continues to remove layers of complexity in infrastructure management. Serverless computing, also known as hosted functions, allows event-driven decoupled application component designs resulting in a high level of component reuse and lower operating costs. Students identify appropriate use-cases for integrating serverless functions into larger solution architectures. Guided lab scenarios give students hands-on practice with designing and implementing serverless application components.

## Vocational Learning Outcomes

This course provides the opportunity for you to achieve the following outcomes:

## 1531X01FWO - Cloud Development and Operations

**VLO 1** Create an appropriate architectural design to support the client`s secure virtual infrastructure and application development needs. (T, A)

**VLO 2** Select and deploy appropriate cloud environments to support a variety of workloads in a rapidly changing business environment. (T, CP)

**VLO 3** Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, A)

**VLO 4** Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, A)

**VLO 5** Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, A)

**VLO 6** Evaluate the full cost implications of various cloud solution options to find an optimal cost/benefit balance for an organization. (T, A)

**VLO 7** Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, A)

**VLO 9** Identify and apply discipline-specific practices that contribute to the local and global community through social responsibility, economic commitment and environmental stewardship. (T, CP)

## 1531X03FWO - Cloud Development and Operations

**VLO 1** Create an appropriate architectural design to support the client`s secure virtual infrastructure and application development needs. (T, A)

**VLO 2** Select and deploy appropriate cloud environments to support a variety of workloads in a rapidly changing business environment. (T, CP)

## 1531X07PAO - Cloud Development and Operations

**VLO 1** Create an appropriate architectural design to support the client`s secure virtual infrastructure and application development needs. (T, A)

**VLO 2** Select and deploy appropriate cloud environments to support a variety of workloads in a rapidly changing business environment. (T, CP)

## 1531X09FAO - Cloud Development and Operations

**VLO 1** Create an appropriate architectural design to support the client`s secure virtual infrastructure and application development needs. (T, A)

**VLO 2** Select and deploy appropriate cloud environments to support a variety of workloads in a rapidly changing business environment. (T, CP)

**VLO 3** Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, A)

**VLO 4** Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, A)

**VLO 5** Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, A)

**VLO 6** Evaluate the full cost implications of various cloud solution options to find an optimal cost/benefit balance for an organization. (T, A)

**VLO 7** Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, A)

**VLO 9** Identify and apply discipline-specific practices that contribute to the local and global community through social responsibility, economic commitment and environmental stewardship. (T, CP)

**Assessment Levels —T**: Taught **A**: Assessed **CP**: Culminating Performance

# Course Learning Outcomes / Elements of Performance

When you have earned credit for this course, you will have demonstrated the ability to:

**1. Describe the characteristics of serverless applications.**

- **1.1** Identify the key components of serverless architecture, including Function as a Service (FaaS) and Backend as a Service (BaaS).
- **1.2** Explain the operational advantages of serverless applications, such as auto-scaling, reduced infrastructure management, and pay-per-use pricing.
- **1.3** Discuss the common use cases for serverless applications, including real-time data processing, event-driven applications, and microservices
- **1.4** Recognize potential limitations and challenges of serverless applications, such as cold starts, statelessness, and vendor lock-in

**2. Distinguish serverless application designs from containerized solution architectures**

- **2.1** Compare the architectural principles of serverless and containerized applications, highlighting differences in deployment, management, and scaling
- **2.2** Describe the role of containers in containerized applications, including the use of Docker and Kubernetes for orchestration.
- **2.3** Analyze the trade-offs between serverless and containerized solutions, considering factors such as cost, flexibility, and complexity
- **2.4** Examine case studies demonstrating the selection of serverless or containerized architectures based on specific application requirements.

**3. Construct complete example applications using only serverless application components.**

- **3.1** Design a serverless application architecture that addresses a given problem or use case.
- **3.2** Describe the role of containers in containerized applications, including the use of Docker and Kubernetes for orchestration
- **3.3** Analyze the trade-offs between serverless and containerized solutions, considering factors such as cost, flexibility, and complexity
- **3.4** Examine case studies demonstrating the selection of serverless or containerized architectures based on specific application requirements

**4. Employ the appropriate integration of serverless functions alongside more traditional application component architectures**

- **4.1** Evaluate scenarios in which serverless functions can complement traditional application components, such as microservices or monoliths
- **4.2** Implement hybrid architectures that combine serverless functions with containerized or virtual machine-based components
- **4.3** Develop strategies for seamless communication between serverless functions and other application components, including API design and event-driven integration
- **4.4** Assess the impact of integrating serverless functions on application maintainability, scalability, and cost-efficiency

**5. Estimate the cost differentials between building applications in serverless, containerized and virtual machine environments**

- **5.1** Analyze the cost structure of serverless, containerized, and virtual machine environments, taking into account factors like resource usage, data storage, and networking
- **5.2** Calculate the Total Cost of Ownership (TCO) for applications in each environment, considering both direct and indirect costs
- **5.3** Identify cost optimization strategies for serverless, containerized, and virtual machine environments, such as autoscaling, right-sizing, and spot instances
- **5.4** Evaluate the cost-effectiveness of each environment for different application scenarios, considering factors like performance, scalability, and complexity.

**6. Evaluate the security and data privacy implications of various serverless application solutions.**

- **6.1** Examine the authentication and authorization mechanisms used in serverless solutions.
- **6.2** Provide recommendations for enhancing the security of serverless applications.
- **6.3** Assess the network security measures in place, including encryption protocols and secure communication channels within serverless architectures.
- **6.4** Identify how monitoring and telemetry can be used to secure serverless applications.

## Learning Resources

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**Required Textbook:**

**Recommended Resources:**

Katzer, Jason (2020). Learning Serverless: Design, Develop, and Deploy with Confidence. Location: https://learning.oreilly.com/library/view/learning-serverless/9781492057000/. Publisher: O'Reilly Media. ISBN: 1492057010, 9781492057017

**Hardware and Software:**

- *Azure Student Subscription*

To access your course list, learning resources and costs, visit Booklist.
Please note that a learner must be registered in a specific course or program to view this information.
Program tuition and fees can be found on the Tuition and Fees Estimator for Full-time Programs page.

# Learning Activities

During this course, you are likely to experience the following activities:

- Lectures and demonstrations
- Hybrid learning activities
- Class discussions
- Presentations
- Assignments
- Quizzes
- Tests

# Pre-defined Evaluation / Earning Credit

The following list provides evidence of this course's learning achievements and the outcomes they validate:

**Assignment(s) (30%)**

Validates Outcomes:  CLO 1, CLO 2, CLO 3, CLO 4, CLO 5, CLO 6

**Final Exam (20%)**

Validates Outcomes:  CLO 1, CLO 2, CLO 3, CLO 4, CLO 5, CLO 6

**Midterm Exam(s) (20%)**

Validates Outcomes:  CLO 1, CLO 2, CLO 3, CLO 6

**Quiz(zes) (20%)**

Validates Outcomes:  CLO 1, CLO 2, CLO 3, CLO 4, CLO 6

**Oral Exam (10%)**

Validates Outcomes:  CLO 1, CLO 2, CLO 3, CLO 4, CLO 5, CLO 6

# Prior Learning Assessment and Recognition

Students who wish to apply for Prior Learning Assessment and Recognition (PLAR) need to demonstrate competency at a post-secondary level in all outlined course learning outcomes. Evidence of learning achievement for PLAR candidates includes:

- Challenge Exam
- Performance Test

# Course Related Information

**Grades**
The minimum passing grade for the course is 50% (D-). Course evaluation is based on a variety of evaluated submissions including quizzes, labs, assignments, oral exam, midterm and final exams; however, to pass the course, students must also achieve a minimum grade of 50% on the Oral Exam, regardless of their overall course average.

# Program Information

## 1531X01FWO - Cloud Development and Operations

**Cloud Usage**
Students in this program may be exposed to Cloud Software through student accounts but they should never use their Credit Card for these services. Algonquin College will not be responsible for a students' utilization of Online Services or Cloud Based Applications.

**Appropriate Use of Generative Artificial Intelligence (GenAI) Tools and AI-Assisted Resources**

In alignment with Algonquin College Policy AA48 (Academic Integrity), students must follow the specific evaluation instructions for each assessment and consult the professor if they are unsure whether GenAI tools are permitted. For the purposes of this course, "GenAI/AI-assisted resources" includes any tool, device, software, browser feature, plug-in, or service that can generate, rewrite, summarize, translate, solve, or provide suggested answers or content (e.g., chatbots, writing/code assistants, AI search tools, and AI-enabled devices such as smart glasses or similar wearable technology). Unless an assessment explicitly allows such tools/resources, students should assume they are **not permitted**. Where use is allowed, students must clearly acknowledge and/or cite that use as directed by the instructor. Submitting AI-generated or AI-rewritten content as one's own work, using AI outside the permitted scope, or failing to acknowledge/cite permitted use may be treated as academic misconduct under AA48.

## 1531X03FWO - Cloud Development and Operations

**Cloud Usage**

Students in this program may be exposed to Cloud Software through student accounts but they should never use their Credit Card for these services. Algonquin College will not be responsible for a students' utilization of Online Services or Cloud Based Applications.

**Appropriate Use of Generative Artificial Intelligence (GenAI) Tools and AI-Assisted Resources**

In alignment with Algonquin College Policy AA48 (Academic Integrity), students must follow the specific evaluation instructions for each assessment and consult the professor if they are unsure whether GenAI tools are permitted. For the purposes of this course, "GenAI/AI-assisted resources" includes any tool, device, software, browser feature, plug-in, or service that can generate, rewrite, summarize, translate, solve, or provide suggested answers or content (e.g., chatbots, writing/code assistants, AI search tools, and AI-enabled devices such as smart glasses or similar wearable technology). Unless an assessment explicitly allows such tools/resources, students should assume they are **not permitted**. Where use is allowed, students must clearly acknowledge and/or cite that use as directed by the instructor. Submitting AI-generated or AI-rewritten content as one's own work, using AI outside the permitted scope, or failing to acknowledge/cite permitted use may be treated as academic misconduct under AA48.

# Department Information

**STUDENT ACADEMIC RESPONSIBILITIES**

Each student is responsible for:

- Knowing the due dates for marked out-of-class assignments.
- Attending all classes and knowing the dates of in-class marked assignments and exercises.
- Arriving to class on time and remaining for the duration of the scheduled class.
- Maintaining a folder of all work done in the course during the semester for validation claims in cases of disagreement with faculty.
- Keeping both paper and electronic copies of all assignments, marked and unmarked, in case papers are lost or go missing.
- Regularly checking both Brightspace announcements as well as one's Algonquin e-mail account for important messages from both professors and college administration.
- Participating in on-line and classroom exercises and activities as required.
- Retaining course outlines for possible future use to support applications for transfer of credit to other educational institutions.

**Harassment/Discrimination/Violence will not be tolerated.** Any form of harassment (sexual, racial, gender or disability-related), discrimination (direct or indirect), or violence, whether involving a professor and a student or amongst students, will not be tolerated on the college premises. Action taken will start with a formal warning and proceed to the full disciplinary actions as outlined in Algonquin College Policies - HR22 and SA07.
Harassment means one or a series of vexatious comment(s) (whether done verbally or through electronic means), or conduct related to one or more of the prohibited grounds that is known or ought reasonably to be known to be unwelcome/unwanted, offensive, intimidating, derogatory or hostile. This may include, but is not limited to: gestures, remarks, jokes, taunting, innuendo, display of offensive materials, offensive graffiti, threats, verbal or physical assault, stalking, slurs, shunning or exclusion related to the prohibited grounds.

For further information, a copy of the official policy statement can be obtained from the Student Association.

**Violation of the Copyright Act**
**General – The Copyright Act** makes it an offence to reproduce or distribute, in whatever format, any part of a publication without the prior written permission of the publisher. For complete details, see the Government of Canada website at http://laws.justice.gc.ca/en/C-42 . Make sure you give it due consideration, before deciding not to purchase a textbook or material required for your course.

**Software Piracy - The Copyright Act** has been updated to include software products. Be sure to carefully read the licensing agreement of any product you purchase or download, and understand the terms and conditions covering its use, installation and distribution (where applicable). Any infringement of licensing agreement makes you liable under the law.

**Disruptive Behaviour** is any conduct, or threatened conduct, that is disruptive to the learning process or that interferes with the well being of other members of the College community. It will not be tolerated. Members of the College community, both students and staff, have the right to learn and work in a secure and productive environment. The College will make every effort to protect that right. Incidents of disruptive behaviour must be reported in writing to the departmental Chair as quickly as possible. The Chair will hold a hearing to review available information and determine any sanctions that will be imposed. Disciplinary hearings can result in penalties ranging from a written warning to expulsion.

For further details, consult the Algonquin College Policies AA32, SA07 and IT01 in your Instaguide.

## College Related Information

**Algonquin College's policies have been developed to ensure the health, safety and security of all students, faculty and staff, and the proper and fair operation of the College as an academic institution and employer. Please refer to the Algonquin College Policies website for the most current policy information available at** http://www.algonquincollege.com/policies/.

**Students are especially encouraged to be aware of the following College expectations**

**Academic Integrity**
Algonquin College is committed to the highest standards of academic integrity, and students are expected to uphold these standards throughout their learning journey with the College and post-graduation. Any academic work submitted by a student is expected to be their own work, unless designated otherwise, all human and technological assisted generated sources must be attributed. Refer to the College's Library web site for attribution support and resources. All students are expected to be familiar with the College's AA48: Academic Integrity policy which outlines student and staff roles and responsibilities, the process for addressing, determining and ruling final academic decisions regarding violations of academic integrity as well as what represents academic misconduct. Students with any questions about academic integrity course expectations including but not limited to use of technology, generative AI and other resources to support completion of coursework, are encouraged to speak to their professor and refer to the course weekly schedule, assessment instructions, course announcements and materials.

**Centre for Accessible Learning (CAL)**
Students with visible and/or non-visible disabilities are encouraged to register with the Centre for Accessible Learning (CAL) in order to be eligible for appropriate learning supports and/or accommodations. Students are strongly encouraged to make an appointment with the Centre for Accessible Learning as early as possible when starting a program. Once your needs are identified, a Letter of Accommodation (LOA) will be issued which you can share with your professors through the CAL Student Portal. If you are a returning student, please ensure that professors are given a copy of your LOA each semester.

**Test Accommodations for Students Registered with CAL**
Students who require testing accommodations will book their tests/exams through the CAL Student Portal. It is important to note the posted test/exam booking deadlines for the term. Test booking deadlines are communicated through CAL Disabilities Counsellors, CAL Testing Services, and other college wide communication. Additional information on the steps to book accommodated tests/exams with CAL Test Services can be found at Step-by-Step Overview.

**Retroactive Accommodations**
Students are expected to meet evaluation and completion deadlines as stated in course outline and weekly schedule documents. In circumstances where evaluation deadlines are missed or student performance has been affected by a temporary, significant and/or unanticipated change in functional ability related to disability (including mental health), interim or retroactive accommodations may be considered. For these instances it is advised you reach out to CAL or your professors as soon as possible. For other situations where deferral of evaluations may be warranted, please consult Algonquin College Policy AA21: Deferred Evaluation.

**Network Credentials and College Email**
Students at Algonquin College are provided with a college email account. This is the address that will be used when the College, your professors, or your fellow students communicate important information about your program or course activities. Your network credentials can be found in the ACSIS portal and you are expected to check your Algonquin email regularly and to use it to send and receive college-related email. Students are required to maintain the privacy of their AC login credentials. Sharing network credentials poses a security risk and is subject to disciplinary action. Your responsibility to protect your login credentials is governed by the Information Security Policy IT01. Support is available through the college Information Technology Service (ITS) at: https://www.algonquincollege.com/its/

**Mental Health and Wellness Supports**

Canada has launched a national Suicide Helpline at www.988.ca and a wide range of mental health and wellness resources for Algonquin students are available at www.algonquincollege.com/SSS/student-health-wellness.

**Student Course Feedback**

Algonquin College invites students to share their course experience by completing a student course feedback survey for each course they take. For further details consult Algonquin College Policy AA25: Student Course Feedback.

**Use of Mobile Devices in Class**

With the proliferation of small, personal mobile devices used for communications and data storage, Algonquin College believes there is a need to address their use during classes and examinations. During classes, the use of such devices unless authorized by your professor can be disruptive and disrespectful to others. During examinations, the use of such devices is generally prohibited unless authorized by your professor. Otherwise use is considered academic dishonesty in the form of cheating. For further details consult Algonquin College Policy AA32: Use of Electronic Devices in Class

**Technology Requirements**

Students are required to have access to a computer and to the internet. There may also be additional technology-related resources required to participate in a course that are not included in the course materials fee, such as headphones, webcams, specialized software, etc. Details on these requirements can be found in the Weekly Schedule document of the course for each course available on Brightspace.

**Transfer of Credit**

It is the student's responsibility to retain course outlines for possible future use to support applications for transfer of credit to other educational institutions.

**Safe Harbour**

In the event of an unexpected major event (pandemic, etc.), your course may have changes that are not reflected in the Course Outline. Should this happen, the Weekly Schedule document will have updated information about your course.